

ALEXANDER STEPHEN MILES**PERSONAL DETAILS**

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PROFILE

I am a data scientist and analyst with 15 years of experience in multiple business sectors: pharmaceuticals, music, transport, research and insurance. I graduated from the University of Warwick with a degree in Mathematics in 2008 and then worked for six years as an analyst in the UK and another six years in Ireland. My duties as an analyst included visualisation, systems management and project management. My duties as a data scientist included machine learning, OCR, classification models, packing algorithms and parameter fitting. I worked for 2 years as a researcher (MSc) developing complex statistical analysis for modelling the immune system. This study was based partly in Ireland and partly as a visiting student in the USA at the Massachusetts Institute of Technology (MIT). My last role was as a data scientist in Ireland until I returned to the UK due to bereavement. My areas of proficiency are databases, mathematical analysis, programming, data manipulation, modelling, optimisation algorithms, predictive analysis, statistics and computational mathematics.

WORK HISTORY**Mar 2025 - Present: Senior Data Analyst at the Construction Industry Training Board (CITB) (U.K.)**

- **Leading Data Migration Extract Transform Load.** I built an automation process transforming HR data for a new Oracle system. This meant a process that had taken 5 months took a fraction of the time and brought the project back on track. This included cleaning, validation and matching. This process was successful and was repeated for Finance.
- **Producing programmed tools to help the business.** I produce tools to help the business, including visualisations of cross-hierarchy mappings (manager, cost centre and department) including visualisations of cross-hierarchy mappings (manager, cost centre and department), which senior management used to build a new organisational structure. Other tools included a scheduling tool to map work to availability and a deduplication tool.
- **Pipeline cleaning of large data sets:** The business had a dataset containing millions of records, including duplicates and outdated or incorrect data. I proposed a Python pipeline solution to clean and fuzzy match the data and provided statistical tests to validate it.

Jun 2022 - Jan 2024: Data Scientist at Ardonagh (Ireland)

- **Building several proof-of-concept projects to support our subsidiary companies.** I built cross-sell and up-sell models using logistic regression and random forest algorithms. These models showed opportunities where current customers could be targeted with new insurance products. I extracted data from websites via scraping or APIs in order to inform marketing. We built a geospatial database to enhance how geographical location related to insurance premiums. We evaluated the market demographics and the business market share of different products by projecting from limited available data. I built OCR techniques to automatically read files to reduce manual data input. These projects helped the subsidiary companies to reduce costs, reach out to more customers and improve the accuracy of premiums.
- **Use of different data science tools.** Work was done in Python with ML libraries (Keras, TensorFlow, PyTorch, scikit-learn) and R. Other tools used included Power BI, SQL and Azure.

Mar 2020 - Jun 2022: Senior Data Intelligence Analyst at Eli Lilly (Ireland).**Apr 2018 - Feb 2020: Data Analyst at Eli Lilly (Ireland).**

- **Reporting on global travel expenses across a multinational company.** My critical role was as lead analyst responsible for reporting and governance of travel and expense data across the global business. I produced regular essential reports for senior consideration (CEO and Monthly Executive Committee) and for corporate projects. Data was extracted from multiple systems, and from this I identified trends in spend, highlighted drivers, built predictive analytics, found opportunities for improvement and suggested metrics to review and amend travel policy. For example, I modelled the optimum terms for our Cognos contract. I used tools such as Tableau, Python, Hadoop and Cognos.
- **Ad hoc programming and troubleshooting of reports across different departments.** In addition to my role in expense analytics, I gave other departments technical assistance on reporting automation and best practice. For

example, I helped with the building of random sampling models for QA purposes.

Oct 2015 - Oct 2017: MSc research student at Maynooth University (Ireland) and visiting student at MIT (USA).
See the education section for more details.

Apr 2014 - Sep 2015: Data Analyst at Cogitare (UK).

Cogitare is a railway infrastructure optimisation consultancy.

- **Upgrading planning optimisation.** The company was hired by Network Rail to coordinate a plan for closing the railway and implementing upgrade works. I was tasked with requirements gathering and the optimisation of different metrics to significantly improve efficiency. The stakeholders approved and adopted our methodology as best practice for future projects.
- **Programming models for clients.** As part of a team, I helped build software to automate the optimisation process of the above project. The model had to return an 'access plan' based on an optimisation algorithm that grouped engineering works. I built the user interface, the data architecture and the packing algorithm. This model received excellent positive feedback from the customers.

Oct 2010 - Apr 2014: Business Analyst at the Performing Rights Society (PRS, UK).

Apr 2009 - Oct 2010: Finance Analyst at Johnson and Johnson (UK).

Sep 2008 - Apr 2009: Technician at Richmond upon Thames College (UK).

Sep 2005 - Jul 2008: BSc student at the University of Warwick (UK). (See the education section for more details.)

EDUCATION

Oct 2015 - Oct 2017: MSc Research Student at Maynooth University, spending one year as a visiting student at MIT.

- **Writing a thesis with an original scientific contribution.** I completed my thesis on 'Mathematical Modelling and Statistical Inference from Immune Response Data' and received my MSc. This thesis was published as a paper (<https://link.springer.com/book/10.1007/978-3-030-57204-4>). The thesis received strong praise from the external examiners (e.g. "a high-quality thesis that presents interesting findings concerning an important question").
- **Mathematical modelling of the immune system.** My main task was to develop a mathematical model for the immune response that was supported by experimental evidence. Using experimental results seen in different publications, I developed creative ways to formulate these results into a mathematical model. This required self-motivated background research to verify that the modelling was consistent with observed results. I fitted models and programmed simulations (Python) to validate assumptions.

Sep 2005 - Jul 2008: BSc Mathematics (2:1) University of Warwick. Modules included: Functional Analysis, Complex Analysis, Partial Differential Equations, Measure Theory, Probability, Statistics, Game Theory, Metric Spaces, Combinatorics, Group Theory, Number Theory, and Computing (MATLAB).

Sep 1998 - Jul 2005: Tiffin School Kingston. A levels: Grade A in Mathematics, Physics and Biology. AS levels: Grade A in Critical Thinking and Grade B in Theatre Studies. GCSE: 9 full courses and 2 short courses.

SKILLS

- Programming Languages: Proficient in Python, VBA, SQL, GML, Cognos. Experience with R, MATLAB, Octave, Java.
- Proficient in Microsoft Office. Experience with Microsoft Azure.
- Azure Fundamentals (Microsoft Certified). Azure Data Scientist Associate (Microsoft Certified). Microsoft Excel Expert (Microsoft Certified). PRINCE2 Foundation qualification in project management (APMG). ITIL Foundation qualification in IT management (AXELOS). Certificate in Machine Learning (Coursera, Stanford University). Specialisation Python FE with honours (Coursera, University of Michigan). Cost and Economics in Pricing Strategy (Coursera, University of Virginia). Data Science Professional Certificate with honours (Coursera, IBM).
- Languages: English (Native), Spanish (Basic).

INTERESTS

Creative writing, swimming, history. I program strategy and puzzle games as a hobby. Store page: <https://store.steampowered.com/search/?developer=Minmaxdog>

Alexander Stephen Miles

C.V.

Full driving licence. No convictions.
REFERENCES AVAILABLE ON REQUEST